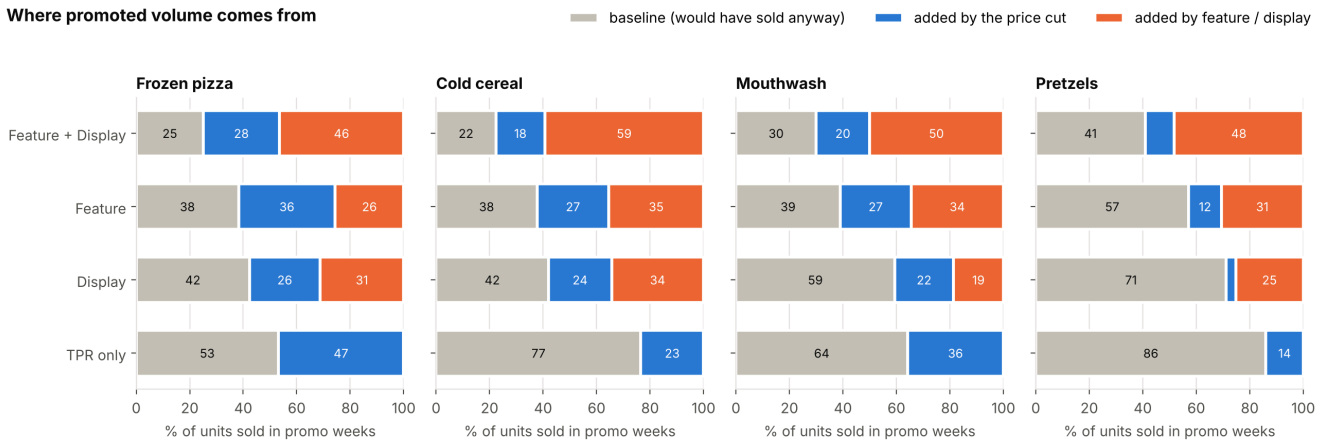
Promotions and pricing in four grocery categories: what pays off

To: Category and Revenue Management teams · **From:** Essoréké Kiheou, Data Science · **Data:** dunnhumby *Breakfast at the Frat*, 77 stores, 55 products, Jan 2009 – Jan 2012

Summary

- **Price cuts without merchandising support lose money.** Shelf-tag-only promotions (TPR) took 22 % of the trade spend; two thirds of that budget paid for units that would have sold anyway. At a 35 % gross margin they cost about **$436k of gross profit** over three years.
- **Feature and display, not the discount, drive incremental volume.** With feature + display, 59–78 % of promoted units are truly incremental, and support brings most of that extra volume. These promotions break even at a gross margin of about 35 %.
- **A weekly forecast that knows the promotion plan is 36 % more accurate than a moving average** (WAPE 29 % vs 45 %) and does not under-forecast promotion weeks, which the moving average misses by 35 %.

1. Where the promotion budget goes



For each promoted week, a model of price and promotion effects estimates the **baseline**, the units the same store would have sold that week at the regular price with no support. The rest is incremental, split between the price cut and feature/display.

- **TPR only:** 53 % (frozen pizza) to 86 % (pretzels) of promoted units are baseline volume sold at a discount. Not one category recovers the cost at normal margins (median break-even margin 46 % to 139 %).
- **Display on pretzels is the best promotion in the data:** a 7 % average discount, +29 % volume, profitable even at a 25 % margin (ROI +0.57 at 35 %).
- **Mouthwash promotions are the hardest to justify**: break-even margins of 42 % to 84 % whatever the mechanic.

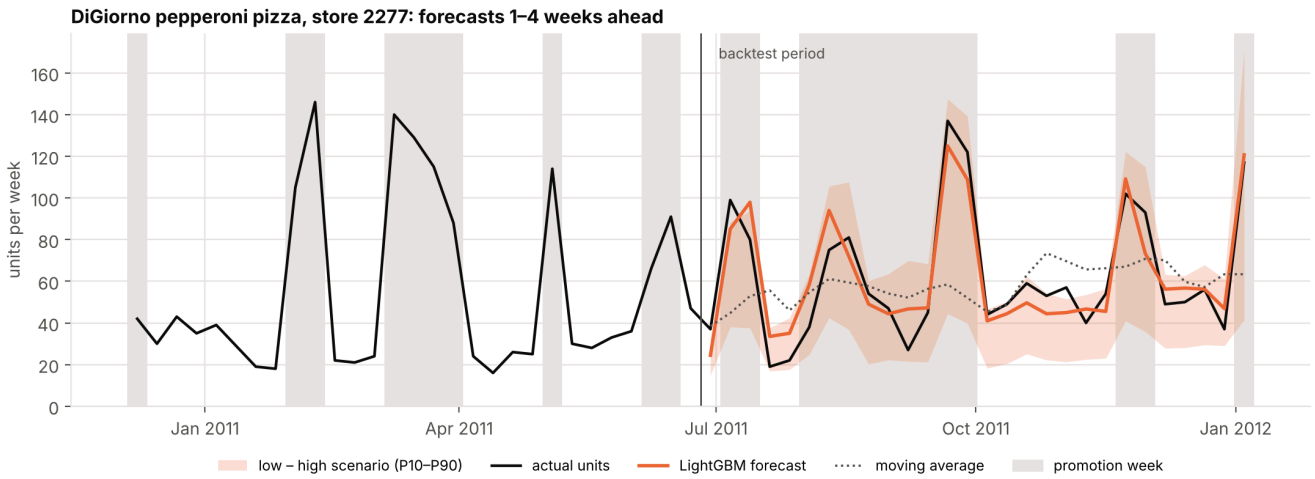
At a 35 % gross margin	Pizza	Cereal	Mouthwash	Pretzels
ROI of TPR only (per $ of discount)	−0.27	−0.60	−0.55	−0.72
ROI of feature + display	−0.02	−0.08	−0.20	−0.02
ROI of display	−0.03	−0.16	−0.33	**+0.57**

2. How price-sensitive each category is

	Pizza	Cereal	Mouthwash	Pretzels
Median own-price elasticity	−2.7	−1.7	−1.2	−1.2
Units after a 10 % price cut	+33 %	+20 %	+14 %	+13 %
Gross margin needed for that cut to add profit	41 %	60 %	83 %	86 %

All 55 products are price-elastic, but only pizza and, to a lesser extent, cereal react enough for a plain price cut to make sense. **Who gains when a leader raises its price:** private label takes most of the switching (DiGiorno +10 % → private-label pizza +8.5 %; Listerine +10 % → private-label mouthwash +5.5 %). **Snyder's discounts mostly cannibalise other Snyder's products**, so product-level results overstate what the brand gains.

3. Forecasting promoted weeks



Example — DiGiorno pepperoni pizza in one store: the forecast follows the promotion peaks that the moving average misses, and the shaded band shows the low and high scenarios. Across all products, backtest over the last 28 weeks, forecasting 1 to 4 weeks ahead with the promotion plan known in advance:

	Same week last year	Moving average	**LightGBM with promo plan**
Error (WAPE)	60 %	45 %	**29 %**
Bias in promotion weeks	−30 %	−35 %	**−2 %**
Bias in normal weeks	+26 %	+21 %	**+2 %**

The forecast also gives a low and a high scenario (P10–P90); they currently contain actual sales 77 % of the time against an 80 % target and should be calibrated before use.

Recommendations

1. **Phase out TPR-only promotions**, starting with pretzels and mouthwash, where they absorb about half of the trade spend. Move the budget to display, which adds more volume per dollar of discount.
2. **Make display with a shallow discount the default mechanic on pretzels.**
3. **Keep feature + display in pizza and cereal only where the funded margin exceeds ~35 %**, and test shallower discounts with the same support: support, not depth, drives the extra volume.
4. **Evaluate every promotion on incremental profit**, counting the discount given on baseline units, not on volume lift.
5. **Plan supply with a forecast that reads the promotion calendar**, and keep the moving average only as a benchmark to detect drift.

Method and limits

Log-log demand models per product with store effects, seasonality, holiday and competitor controls; counterfactual baselines from the same models; rolling-origin backtest for the forecast. 59–84 % of price changes are chain-wide, so elasticities with week fixed effects (smaller) are reported as a lower bound. **Not included:** circular and display fees (they would lower every supported ROI), real margins (results are given as break-even margins and scenarios), cannibalisation and store switching in the ROI. Code, tests and full results: github.com/essoreke-analyse/promo-pricing-analytics.